Suresh Raghu

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Education

Indian Institute of Technology, Madras — Tamil Nadu, India
B.Sc. in Programming & Data Science — *Oct. 2020 – Jun. 2026*

Vellore Institute of Technology, Bhopal — Madhya Pradesh, India
B.Tech in Computer Science and Engineering — *Oct. 2020 – Jun. 2024*

Research Interests

LLM Agents & Scientific Agents | Reinforcement Learning & Tool Use | Uncertainty Quantification | Multimodal Reasoning

Publications & Pre-Prints

Proper Scoring Rules for Agentic Uncertainty Quantification.
Suresh Raghu*, *Satwik Pandey**, *Shashwat Pandey* — *(Under Review)*
arXiv:2605.24756 | **Accepted (poster) at CTB & FAGEN Workshops @ ICML 2026**

- Developed **strictly proper scoring rules for uncertainty over LM-agent trajectories**, including a **censored-trace formulation** for incomplete executions; showed that standard trajectory-level adaptations of **ECE and Brier score** need not remain strictly proper in agentic settings.

SELFDOUBT: Uncertainty Quantification for Reasoning LLMs via the Hedge-to-Verify Ratio.
*Satwik Pandey**, ***Suresh Raghu****, *Shashwat Pandey* — *(Under Review)*
arXiv:2604.06389 | **Accepted (poster) at FAGEN Workshop @ ICML 2026**

- Proposed the **Hedge-to-Verify Ratio (HVR)**, a single-pass $\mathcal{O}(1)$ uncertainty signal that scores a reasoning trace by how much it hedges versus self-verifies. Traces with no hedging language are correct **96.1% of the time** (at 25.4% coverage), and fusing HVR with the model's verbalized confidence beats sampling-based Semantic Entropy on discrimination ($p = 0.001$) at **10× lower inference cost**, across 7 models and 3 benchmarks.

Don't Blink: Evidence Collapse during Multimodal Reasoning.
Suresh Raghu*, *Satwik Pandey** — * Equal contribution
arXiv:2604.04207 | **Accepted (poster) at Sci-FM & Actionable Interpretability Workshops @ COLM 2026**

- Identified **"evidence collapse"**, a universal (**9/9** model×dataset cells) decay of visual grounding during VLM reasoning that text-only entropy cannot detect, and designed a task-conditional **vision veto** that cuts selective risk by up to **1.9 pp** at 90% coverage on MathVista, HallusionBench, and MMMU_Pro.

Research & Engineering Experience

Carnegie Mellon University – Language Technologies Institute — Remote
Independent Research Collaborator – LLM Agents for Scientific Discovery — *July 2026 – Present*

- **Scientific Agent Harnesses:** Audited and redesigned **SMDD-Bench** agents for long-horizon drug design, adding structured state, evaluator-aligned checks, and task-bound **RDKit, ADMET-AI, and Boltz2** tools; improved **Qwen3.5-9B** lead-optimization success from **16.3% to 51.0%**.
- **RL Environment for Scientific Agents:** Building a **Harbor + Prime Env** environment for reproducible agent rollouts with budgeted scientific tools, evaluator-backed rewards, and trajectory logging for training open-source LLM agents.

VFS Global — New Delhi, India
Senior Manager – AI (Founding Lead, AI Research Engineering) — *May 2024 – Present*

- **Adaptive Query Routing:** Architected the document-extraction pipeline around an adaptive router that sends deterministic documents to lightweight parsers and ambiguous ones to reasoning VLMs, reaching **99.1% field-level extraction accuracy**.
- **Uncertainty-Gated Extraction Cascade:** Used a cheap token-logit trigger to run **Semantic Entropy** only on suspect outputs, escalating flagged cases to a larger LLM and cutting critical extraction errors by **87%**.
- **Biometric Compliance Pipeline:** Proposed and built a two-stage visa photo-verification system consisting of a cheap image-quality gate followed by a VLM stage for pose and occlusion checks, reducing cases requiring manual intervention by **78%**.